Math Natural Language Inference: this should be easy!

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Paper Github

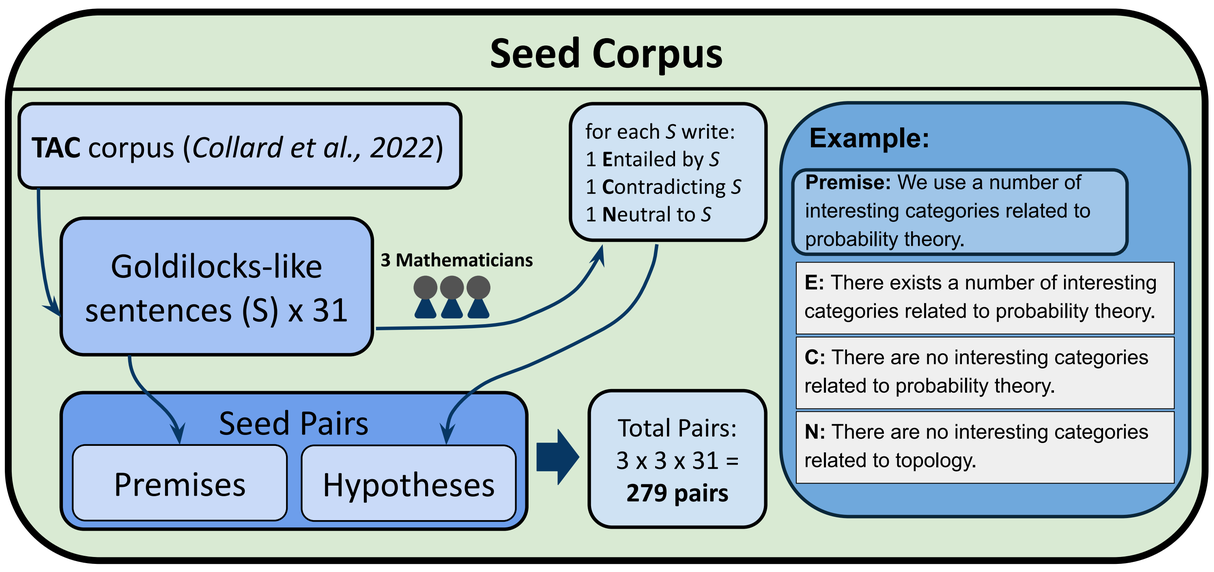
Can LLMs be reliable constructors and annotators of Math NLI corpora?

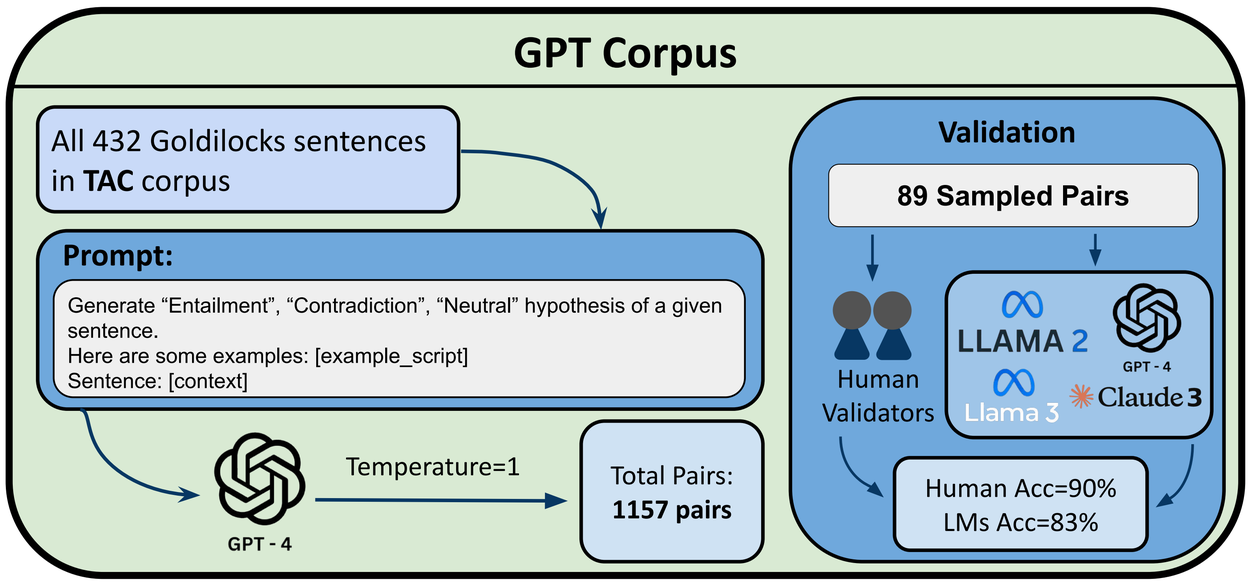
- How well do LLMs perform on a Math NLI corpus annotated by mathematicians?
- Are there common features to the errors which they make?
- How good is a Math NLI corpus annotated entirely by LLMs?
- Are LLMs more unanimous on human-written corpora or on corpora generated by LLMs themselves?

Notation for LLMs

GPT4	GPT4	**Mistral**	Mistral-largel
L2	Llama-2-70B	**Q2**	Qwen2-72B-Instruct
L3	Llama-3-70B	**DS**	Deepseek-llm-67b-chat
L3.1	Llama-3.1-70B-Instruct	**Ge2**	Gemma-2-27b-it
C3	Claude-3		

Corpus Generation





Math NLI Experiments

Seed Corpus

		GPT4	L2	L3	C3	Mistral	L3.1	Q2	Mixtral	DS	Ge2
C	p	82.9	90.5	70.9	91.8	79.8	88.8	87.3	75.4	**92.7**	85.1
	r	**98.9**	61.3	96.8	95.7	97.8	93.5	95.7	**98.9**	81.7	92.5
	f1	90.2	73.1	81.8	**93.7**	87.9	91.1	91.3	85.6	86.9	88.7
E	p	90.1	73.9	85.4	**93.5**	89.8	83.8	80.8	86.3	82.0	82.8
	r	**97.8**	88.2	94.6	92.5	84.9	89.2	90.3	88.2	78.5	82.8
	f1	**93.8**	80.4	89.8	93.0	87.3	86.5	85.3	87.2	80.2	82.8
N	p	**95.5**	56.2	91.8	87.8	81.8	81.7	84.7	85.5	67.6	75.3
	r	68.8	63.4	48.4	**84.9**	67.7	72.0	65.6	57.0	78.5	68.8
	f1	80.0	59.6	63.4	**86.3**	74.1	76.6	73.9	68.4	72.6	71.9
acc		88.5	71.0	79.9	**91.0**	83.5	84.9	83.9	81.4	79.6	81.4
avg	p	89.5	73.5	82.7	**91.0**	83.8	84.8	84.2	82.4	80.8	81.1

(a) GPT4

Gold↓	C	E	N
C	98.9	.0	1.1
E	.0	97.8	2.2
N	20.4	10.8	68.8

(b) Llama2

	C	E	N
C	61.3	1.1	37.6
E	.0	88.2	11.8
N	6.5	30.1	63.4

(c) Llama3

	C	E	N
C	96.8	.0	3.2
E	4.3	94.6	1.1
N	35.5	16.1	48.4

(d) Claude3

	C	E	N
C	95.7	.0	4.3
E	.0	92.5	7.5
N	8.6	6.5	84.9

(e) Mistral

	C	E	N
C	97.8	.0	2.2
E	2.2	84.9	12.9
N	22.6	9.7	67.7

(f) Llama3.1

Gold↓	C	E	N
C	93.5	.0	6.5
E	1.1	89.2	9.7
N	10.8	17.2	72.0

(g) Qwen2

	C	E	N
C	95.7	.0	4.3
E	2.2	90.3	7.5
N	12.0	21.7	66.3

(h) Mixtral

	C	E	N
C	98.9	.0	1.1
E	3.2	88.2	8.6
N	29.0	14.0	57.0

(i) DeepSeek

	C	E	N
C	81.7	.0	18.3
E	2.2	78.5	19.4
N	4.3	17.2	78.5

(j) Gemma2

	C	E	N
C	92.5	1.1	6.5
E	1.1	82.8	16.1
N	15.1	16.1	68.8

GPT corpus

		GPT4	L2	L3	C3	L3.1	Q2	Mixtral	DS	Ge2
C	p	**100.0**	**100.0**	**100.0**	96.6	96.6	96.6	96.7	**100.0**	**100.0**
	r	96.7	53.3	96.7	93.3	93.3	93.3	96.7	83.3	**100.0**
	f1	98.3	69.6	98.3	94.9	94.9	94.9	96.7	90.9	**100.0**
E	p	70.7	57.7	66.7	71.8	65.2	75.0	**76.3**	75.0	73.0
	r	96.7	**100.0**	**100.0**	93.3	**100.0**	**100.0**	96.7	90.0	90.0
	f1	81.7	73.2	80.0	81.2	78.9	**85.7**	85.3	81.8	80.6
N	p	89.5	33.3	**93.3**	85.7	85.7	90.0	90.5	71.4	86.4
	r	58.6	24.1	48.3	62.1	41.4	62.1	65.5	**69.0**	65.5
	f1	70.8	28.0	63.6	72.0	55.8	73.5	**76.0**	70.2	74.5
acc		84.3	59.6	82.0	83.1	78.7	85.4	**86.5**	80.9	85.4
avg	p	86.7	64.0	86.6	84.7	82.5	87.2	**87.8**	82.3	86.4
	r	84.3	59.6	82.0	83.1	78.7	85.4	**86.5**	80.9	85.4

(a) GPT-4

Gold↓	C	E	N
C	96.7	.0	3.3
E	.0	96.7	3.3
N	.0	41.4	58.6

(b) Llama 2

	C	E	N
C	53.3	.0	46.7
E	.0	100.0	.0
N	.0	75.9	24.1

(c) Llama 3

	C	E	N
C	96.7	.0	3.3
E	.0	100.0	.0
N	.0	51.7	48.3

(d) Claude 3

	C	E	N
C	93.3	3.3	3.3
E	.0	93.3	6.7
N	3.4	34.5	62.1

(e) Llama 3.1

	C	E	N
C	93.3	.0	6.7
E	.0	100.0	.0
N	3.4	55.2	41.4

(f) Qwen2

	C	E	N
C	93.3	.0	6.7
E	.0	100.0	.0
N	3.4	34.5	62.1

(g) Mixtral

	C	E	N
C	96.7	.0	3.3
E	.0	96.7	3.3
N	3.4	31.0	65.5

(h) Deepseek

	C	E	N
C	83.3	.0	16.7
E	.0	90.0	10.0
N	.0	31.0	69.0

(i) Gemma2

	C	E	N
C	100.0	.0	.0
E	.0	90.0	10.0
N	.0	34.5	65.5

Patterns we found in flawed reasoning

Ignored context

P: In the nilpotent case, this nerve is known to be a Kan complex.

H: This nerve is not known to be a Kan complex.

LLMs discarded the specific context (*the nilpotent case*) and compared the matrix sentences leading to a contradiction, instead of a neutral label.

Unknown Math Concepts

P: This paper proposes a recursive definition of V-n-categories and their morphisms.

H: This paper proposes a definition of V categories.

Sometimes one must know the concepts involved. Here, only if we know "V-n-categories" are "V-categories" can we then decide on entailment.

Lexical Ambiguity

P: The axioms resemble those for monoidal Abelian categories.

H: The axioms are the ones of monoidal Abelian categories.

There is lexical ambiguity, for example, with the verb "*resemble*" which means "is almost equal" or "it looks similar to something else, but it is ***NOT*** the same as".

Vague Quantifiers

P: We worked through numerous examples to demonstrate the power of these notions.

H: We worked through two examples to demonstrate the power of these notions.

Problems with vague predicates like "*numerous*" here. In this case, our mathematicians agreed that numerous examples should entail two examples, but LLMs did not.

Naming of Math Entities

There is a problem with naming mathematical entities, e.g. "*group B*" vs. "*group C*" if this is only used as a generic name, as an α-variant, then the difference between B and C doesn't matter. But many times we are talking about different groups.

Conclusions

- **Math NLI is not easier than ordinary NLI for LLMs.** Despite exhibiting impressive capabilities in complex mathematical problem-solving, LLMs surprisingly struggle with basic logical reasoning and NLI tasks in mathematics.
- **For LLMs, the order of difficulty is C < E < N.** In human data, C is never the easiest. LLMs frequently identify N pairs in the GPT corpus as E. However, identifying C (contradiction) pairs within the GPT corpus is easier than in the seed corpus.
- The pairs created by humans (the seed corpus of NLI pairs) showed **more evaluation variation** from the LLMs than the corpus created by GPT4.

Funded by a Shanghai Pujiang Program (22PJC063) awarded to Hai Hu.

Do Large Language Models Understand Conversational Implicatures — A case study with a Chinese sitcom

Shisen Yue, Siyuan Song, Xinyuan Cheng, Hai Hu
Shanghai Jiao Tong University

Paper and data:

Contact:
hu.hai@sjtu.edu.cn

Example entry in the dataset (from 武林外传)

Context (target sentece) & Question	Choices	Condition
小郭:知道这意味着什么吗？以后再敢胡来，就不光是挨顿打的事了。 Xiaoguo: Do you know what this means? If you mess around again, it won't just be about getting a beating. *薇*:烦死了，在家就是听爹娘罗嗦，好不容易溜出来，倒听你罗嗦。 Wei: You are so annoying. At home, I have to listen to my parents' nagging. Finally I sneaked out; now I have to listen to your nagging. *小郭*:什么？你是溜出来的？ Xiaoguo: What? You sneaked out? *薇*:我说的那个溜，是溜冰的溜啊，现在京城都在下雪，满地都是冰，我是穿着冰□，溜出来的。 Wei: When I said "sneak out", I meant ice skating. It's snowing in the capital now, the ground is covered in ice. I wore ice skates and slated out. (NB: 溜 can mean both "sneak out" and "ice-skate") Question: What is implied by the target sentence?	A. 薇发现自己说溜出来说漏嘴了，就找补说是溜冰出来的。Wei realized she accidentally slipped up by saying she sneaked out and tried to cover it up by saying she meant ice skating.	Pragmatic (Correct)
	B. 薇是从京城的家里溜冰出来的。Wei sneaked out from her home in the capital to go ice skating.	Semantic
	C. 薇怕下雪冷，所以离开了京城。Wei was afraid of the cold due to the snow, so she left the capital.	Distractor #1
	D. 薇很喜欢溜冰。Wei really enjoys ice skating.	Distractor #2

Gricean maxims annotation

Gricean Maxims	Submaxims & examples (Compliance or Violation)
Quantity	Do not say what you believe is false. Do not say that for which you lack adequate evidence
Quality	Make your contribution as informative as is required. Do not make your contribution more informative than is required.
Relevance	Be relevant
Manner	Avoid ambiguity Be orderly

Meta data

of question: 200
- quality: 76
- quantity: 33
- relation: 71
- manner: 62

Avg. turns: 6.80
Avg. token number in utterances: 29.27
Avg. token number in choices: 15.08

Related works that evaluate LLMs' pragmatic understanding

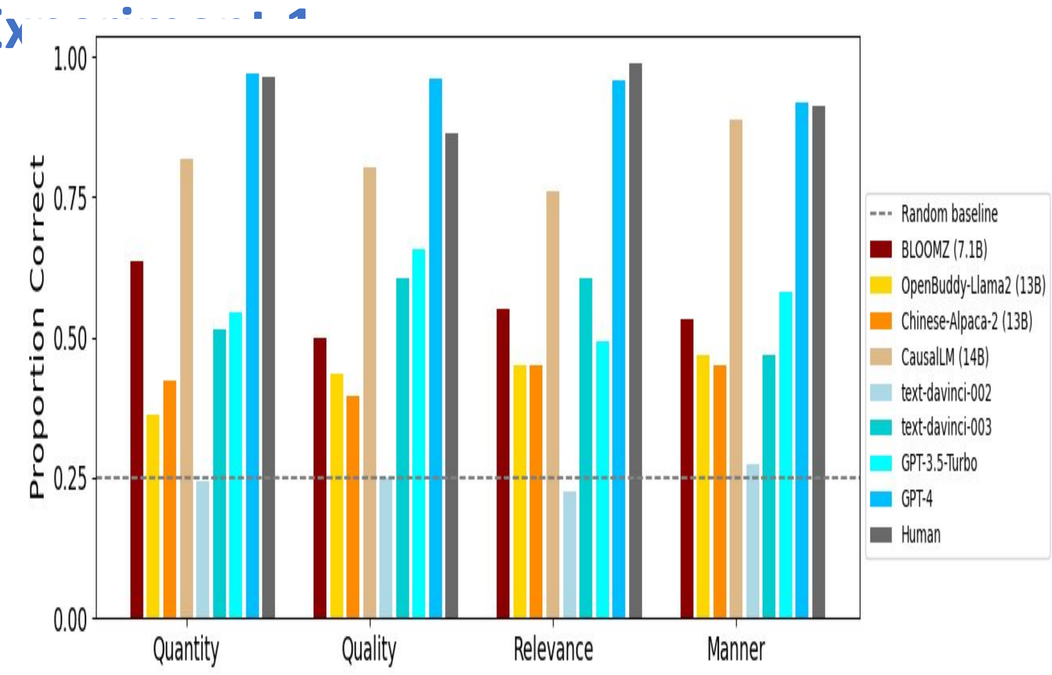
Researchers	Dataset	Context form	Theoretical framework	Language
Hu et al. (2023)	Pragmega	multi-turn dialogues	Theory of Mind	English
Ruis et al. (2022)	LUD WIG	single-turn dialogues	-	English

No research on LLMs' pragmatic understanding in Chinese with a linguistic focus (Gricean cooperative principle)

Experimental design

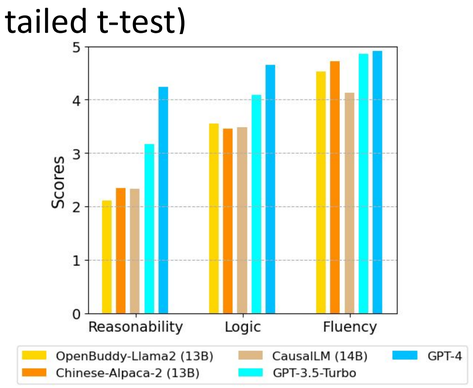
- Experiment 1 requires LLMs to complete multiple-choice questions（Answers are collected by next token prediction for open-source models and free generation for close-source models)
- Experiment 2 requires LLMs to explain implicatures in the dialogue and invites human participants to evaluate their responses from three dimensions (reasonability, logic and fluency)

Results for Experiment 1



- GPT-4 (94%) achieves human-level accuracy (93.1%); others fall behind.
- LLMs' performance do not vary by question type (Gricean maxims violated in the dialogues; $p = 0.802 > 0.05$ by a two tailed t-test)

Results for Experiment 2



Human eval metrics:
1. Fluency: fluent language
2. Logic: logically coherent?
3. Reasonability: truly understands the implicature
   a. Only GPT4 scored high
   b. GPT-3.5-Turbo and open-source models have low score → explanations not convincing to humans

Dialogue example	GPT-4 generated (bad)	CausalLM generated (bad)
老白:啧，说了你也不知道，只要四大□老都在，谁都别想活塌实了 掌柜:有那么恐怖吗 老白:东□老，以前是六扇□的总顾问，眼睛一眯，就知道花盘上的瓜子数 掌柜:咋不去当帐房呢 请根据以上情景判断老白说的"..."有什么言外之意，并解释。	老白的话"东□老，以前是六扇□的总顾问，眼睛一眯，就知道花盘上的瓜子数"的言外之意是，东长老是一个非常细心且观察力极强的人，他的洞察力和判断力非常出色，甚至可以在眼睛一眯的瞬间就知道花盘上的瓜子数。这也暗示了他的智慧和能力，以及他在六扇门的重要地位。同时，这也表明他是一个不容忽视的角色，他的存在使得其他人无法安心。	老白这句话的言外之意是，东□老是一位非常有智慧和洞察力的人，他以前在六扇门担任总顾问，所以他对花盘上的瓜子数有独特的见解。他暗示，如果四大老都在，东□老就能发挥自己的作用。这句话还暗示，东□老有许多秘密和机密，他掌管着许多事情。

maxim of **quantity** (Engelhardt et al., 2006; Okanda et al., 2015; Rubio-Fernandez, 2019)

- Maxims make no difference in LLMs' accuracy in our

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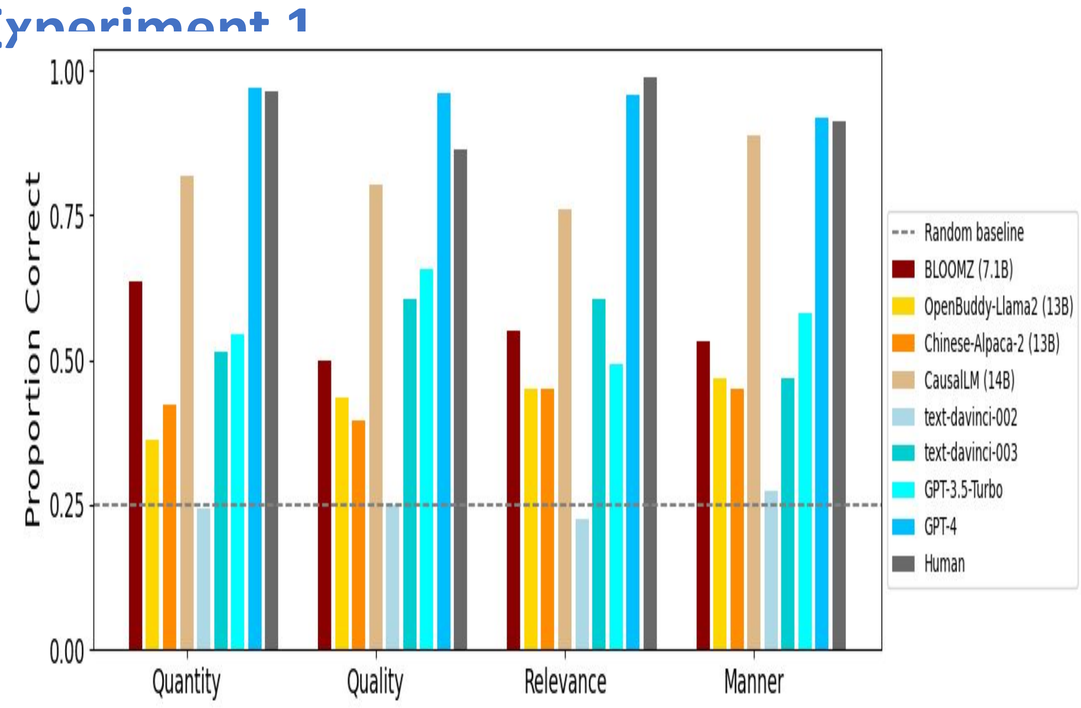

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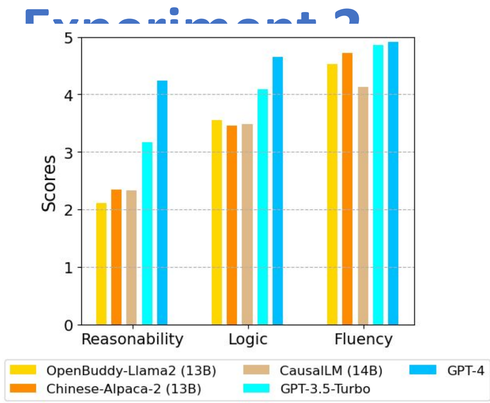

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Avg. token number in responses:

GPT-4: 114.44
GPT-3.5-Turbo: 125.41
Chinese-Alpaca-2: 156.19
CausalLM: 147.41
Openbuddy-Llama2: 153.56

The number of token generated is negatively correlated with the quality of LLMs' response

Discussion

- Can LLMs understand conversational implicatures as humans do?
  - Multiple-choice: GPT-4 on par with ≈ human; but still imperfect in explanation. The results of CausalLM also represent an **unbalanced** performance.
- Do LLMs and humans process pragmatic cues in the same way?
  - Human listeners are less sensitive to violations of maxim of **quantity** (Engelhardt et al., 2006; Okanda et al., 2015; Rubio-Fernandez, 2019)
  - Maxims make no difference in LLMs' accuracy in our study

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Grice an Maxi ms	Submaxims & judgment for the example sentence (Compliance or Violation)
Quant ity	Do not say what you believe is false.
	Do not say that for which you lack adequate evidence
Qualit y	Make your contribution as informative as is required.
	Do not make your contribution more informative than is required.
Relav ance	Be relevant
Mann er	Avoid obscurity of expression.
	Avoid ambiguity
	Be brief
	Be orderly

Hu et al. (2023)	PragMega	multi-turn dialogues	compares probabilities on four choices (A,B,C,D)
Ruis et al. (2022)	LUDWIG	single-turn dialogues	compares probabilities on two analysis (yes or no)

Experiment 1	LLMs do multiple choice questions	prompt_lead <Dialogue> <Four choices>		
Experiment 2	LLMs explain implicatures; human participants rate their responses	prompt_lead <Dialogue> 请		

GPT-4	Human	Causal LM (14B)	text-davinci-003	GPT-3.5-Turbo	BLOOMZ (7.1B)	OpenBuddy-Llama2 (13B)	Chinese-Alpaca-2 (13B)	text-davinci-002